How Much Does a Parity Experiment Prove?

Poorva Barve
Independent Researcher
Seattle, USA
poorvabarve@gmail.com

***Abstract*—Harbor, the evaluation framework behind Terminal-Bench, checks benchmark ports by comparing run scores from the original harness and the port. A port passes when the score ranges overlap. With three runs per side, exchangeable scores under parity, and no ties, the rule flags a perfect port 10 percent of the time. Adding runs makes it less likely to detect a fixed gap. In 58 published comparisons with three runs per side, 57 pass range overlap and only 22 establish equivalence within 3 points. Under normal noise at the median observed variability and a true gap of zero, an equivalence test needs about seven runs per side for 80 percent power at that margin. In 14 of 27 adapters, the two sides disagree on individual tasks more than shared pass probabilities predict at an unadjusted 5 percent threshold. Mean-gap tests find no statistically detectable difference in those adapters. With ten runs per side, range overlap detects a simulated defect that breaks 10 percent of tasks in 36 percent of cases. The pooled and stratified tests each detect it in 86 percent. I propose a parity protocol and provide a verdict tool.**

***Index Terms*—agent evaluation, benchmark portability, equivalence testing, reproducibility**

I. Introduction

Two adapters in the Harbor repository illustrate the problem. For USACO the original harness scored 39.5, 40.5 and 41.0 in three runs and the Harbor port scored 45.7, 42.1 and 41.1. The ranges do not overlap, by 0.1 point, so under the matching criterion of Harbor's contributor guide [1] the port does not match. For RexBench (12 tasks) the original harness scored 33.33 in every run and the port scored 16.67, 58.33 and 50.0. The ranges overlap, so the port passes, with an 8.3-point gap between the means. The same rule gives opposite verdicts, and the comparison it rejects is the one whose data say more about closeness.

Harbor is the evaluation framework behind Terminal-Bench [2]. Existing benchmarks are ported to it through adapters, and the ports support the maintainers' cross-benchmark comparisons and curated Harbor-Index task set [3]. If a port changes scores, it can change conclusions about which agents perform best. The contributor guide requires a parity experiment and treats overlapping run-score ranges as a match [1]. The maintainers' paper asks that the means agree within a margin consistent with their standard errors [3]. Under the guide's rule, adding runs makes a fixed gap harder to detect. This raises the question of how much evidence a pass provides about the size of a possible gap.

I ask three questions. RQ1: what does the range-overlap rule guarantee? RQ2: what do the published runs establish? RQ3: what would a better check cost? My contributions are the rule's error analysis, an audit under a documented metric-selection rule, and run-count and per-task guidance supported by defect simulations. I use public parity files, per-task outcomes and the maintainers' paper; no benchmark was rerun.[1] A failed comparison does not establish an adapter bug; noise in agent evaluations is already documented [4]–[6].

II. Background: Harbor parity experiments

An adapter lets Harbor run an existing benchmark. Before it is merged it goes through code review and a parity experiment: its author runs the same agent and model on the original harness and on the port, k times each with $k = 3$ by default, and reports each side as mean $\pm$ sample standard error of the mean (SEM) [3]; the guide requires at least two runs per side and prefers three or more [1]. The guide asks that the two sides' scores be "statistically indistinguishable" and states the matching criterion as range overlap: $\max(A) \geq \min(B)$ and $\max(B) \geq \min(A)$ for the two lists of run scores A and B. Non-overlap is to be treated as a systematic adaptation error until evidence indicates otherwise, and the parity step is complete when runs on both sides overlap and any remaining gap is documented, with evidence, as sampling noise [1]. The paper states the check twice in other words: the means "agree within a margin consistent with their standard errors" (D.4.2) and the port "must confirm statistical equivalence with the original benchmark within sample SEM" (D.7) [3].

The snapshot is Harbor commit `dc933be` (10 October 2026). Its 84 parity files hold 121 experiments. Eighty-two compare an original harness with its port; 61 have three runs on both sides, and 58 remain after dropping two whose run lists are identical on both sides and one whose metric is not on a 0-to-100 scale. These 58 form the main analysis set. The overlap rule and debugging playbook entered the guide together on 19 April 2026 [7]. The recorded dates of 51 of the 58 comparisons precede that change. The earlier guide asked for comparable averages. I apply the current rule retrospectively to all 58 to characterize its decisions on the recorded scores. Historical reviews are outside the scope of that comparison. From the Hugging Face dataset I recovered per-task result files for 49 adapters, 44 with both sides [8].

[1]Data, scripts, analysis plan with dated amendments, per-task parsers, number-to-script map and verdict tool: https://github.com/barveP/benchmark-port-parity-study.

TABLE I
RULES AT $k = 3$ UNDER NORMAL NOISE. THE LAST COLUMN GIVES DETECTION OF A 2 SD GAP BY A t-STATISTIC CALIBRATED TO EACH RULE'S FALSE-ALARM RATE (FA).

rule	false alarm	1 SD	2 SD	t at same FA
range overlap (Harbor)	0.10	0.26	0.62	0.64
(S1) SEM bars overlap	0.25	0.50	0.86	0.86
(S2) gap within difference SE	0.37	0.63	0.92	0.92
(S3) gap within one SEM	0.44	0.68	0.94	0.94
pooled t-test, 5%	0.05	0.16	0.46	0.46
Welch t-test, 5%	0.03	0.12	0.37	0.38

III. RQ1: WHAT THE RULE GUARANTEES

Closed form. Two sides produce k independent run scores each. Under a perfect port the $2k$ scores are exchangeable, so every ordering is equally likely. The ranges fail to overlap exactly when all k scores of one side exceed all k of the other, which happens in $2(k!)^2$ of the $(2k)!$ orderings, so $\Pr(\text{mismatch} \mid \text{perfect port}) = 2(k!)^2/(2k)!$, that is $1/3$, $1/10$, $1/35$ and $1/126$ at $k = 2, 3, 4, 5$, for any continuous noise distribution. When scores are pass counts that can tie, ties count as overlap and the rate is lower: 4 to 9% at $k = 3$ over a grid of 10 to 1,000 tasks and base rates 0.1 to 0.5.

Other rules. Table I compares six rules at $k = 3$ under normal noise with a common run-to-run standard deviation (SD): range overlap; mean $\pm$ SEM bars overlap (S1) and gap within the standard error of the difference (S2), the two readings of the paper's D.4.2 wording; gap within the larger of the two SEMs (S3), the literal reading of D.7; and pooled and Welch t-tests at 5%. The last column gives the detection of a t-statistic tuned to each rule's own false-alarm rate. S2 and the pooled t-test are one statistic at different thresholds, and S1, S3 and the Welch test are close to it; range overlap is a different statistic, but at its own false-alarm rate it loses only two points of detection (0.62 against 0.64). Inefficiency is not its problem; its problems are structural.

Three properties. First, without ties the false-alarm rate is set by k alone (Fig. 1(a)). Second, for a fixed gap detection cannot increase with more runs. Non-overlap with $k + 1$ runs per side implies non-overlap with the first k. For a 2 SD gap, overlap detection falls from 0.78 at $k = 2$ to 0.09 at $k = 10$, while the pooled t-test's rises from 0.22 to 0.99 (Fig. 1(b)). Third, greater common noise makes a fixed gap harder to detect with both range overlap and the pooled t-test. For a 2-point gap at $k = 3$, the overlap pass probability rises from 0.02 at an SD of 0.5 points to 0.68 at the median observed SD of 1.7 points and approaches 0.90 as noise grows. In the same setting, greater noise makes equivalence harder to establish by TOST (Fig. 1(c)). Simulations using per-task pass probabilities for 30 adapters give a median overlap false-alarm rate of 0.07 with ties.

What a pass means. With three runs per side and exchangeable scores under parity, a pass is a failure to reject with a difference test whose false-alarm rate is at most 10%. It never means that the port is within any stated distance of the original: under a perfect port with $k = 3$ the smallest margin within which equivalence can be shown is 2.3 run SDs on average, which is 3.9 points at the median SD of 1.7. Read literally, the paper's wordings are stricter than the guide's rule: they flag 25 to 44% of perfect ports (Table I).

IV. RQ2: WHAT THE PUBLISHED RUNS ESTABLISH

Method. I built one comparison per experiment under a rule written down before the analysis: which metric row is the main one (the row matching the paper's tables, else the headline metric, else the first point-scale row), which rows are excluded, and how units are handled. Six amendments were logged with date and reason; two refined the rule before any count was computed, and the one made after the first counts left the within-5 count at 36 of 58. For each comparison I computed the gap (Harbor minus original), the pooled-t 90% confidence interval, and the *smallest provable margin*: the distance from zero to the farther end of that interval. Equivalence within $\pm m$ is shown when the whole 90% interval lies inside $\pm m$, the two-one-sided-tests (TOST) procedure at the 5% level [9], [10].

Results. The ports look close. Over the 58 comparisons the median absolute gap is 0.67 points, the median run-to-run SD 1.71 points, and no gap is distinguishable from zero at the 95% level. The evidence that they are close is thin: the quartiles of the smallest provable margin are 2.4, 4.4 and 6.0 points. Equivalence within 2, 3 and 5 points can be shown for 7, 22 and 36 of the 58, and one in four cannot show it even within 6 points (Fig. 2).

The rule's verdicts. Fifty-seven comparisons pass range overlap; 21 of them cannot show equivalence within 5 points and 35 cannot within 3. The rule passes the four largest gaps: Frontier-CS at -15.6 points (a 10-task subset with a run SD of 16 points), RexBench at $+8.3$, PIXIU at -4.9 and GDB at $+4.6$, whose smallest provable margins are 44.2, 35.5, 20.3 and 11.6 points. The one failure, USACO, misses overlap by 0.1 point and has a smallest provable margin of 5.8 points, which 18 of the 57 passes exceed. The paper's SEM wordings would have failed 4, 12 and 18 of the 58 under readings S1, S2 and S3.

Selection. Under independent comparisons of perfect ports, range overlap would flag 5.8 on average without ties or 3.9 at the median modeled rate with ties. The chance of at most one failure is 0.017 and 0.089, respectively. USACO, the only failure, predates the overlap rule and its debugging playbook. The earlier comparable-averages requirement could have favored publication of close results. The records leave run selection unresolved, and the strength of the evidence depends on the noise model. Published pass rates cannot validate the rule. Section III characterizes its error rates under specified noise models.

Stability. Counts depend on the margin and interval method. A 0.1-point shift moves the within-5 count between 34 and 37, and Welch intervals give 31. The quartiles provide a more stable summary. Subset comparisons (22 of the 58) are noisier than full-set ones (median run SD 2.3 against 1.5 points).

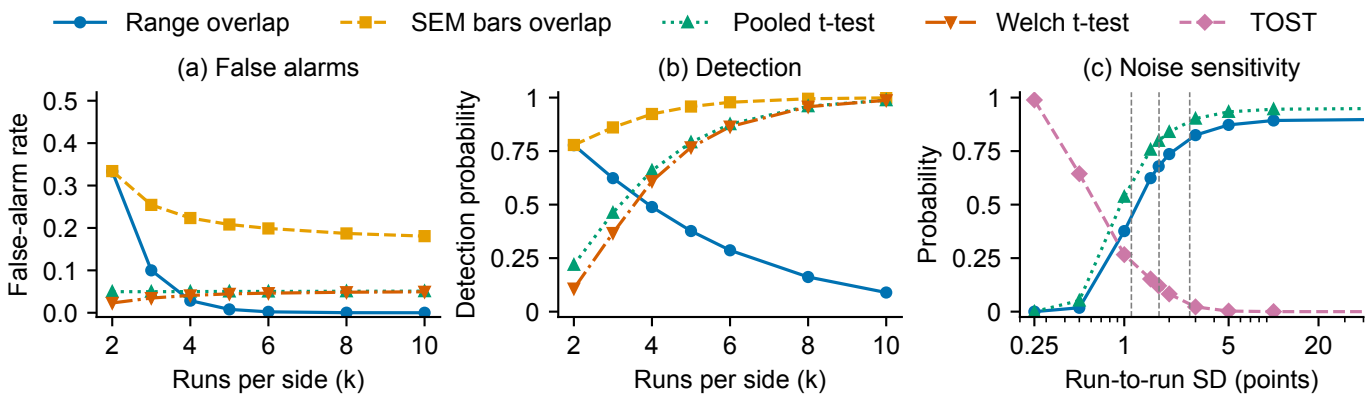


Fig. 1. Normal run-to-run noise. (a) False-alarm rate and (b) detection of a 2 SD gap as runs per side (k) increase. Both t-tests use the 5% level. (c) For a true 2-point gap at $k = 3$, the curves show the probabilities that range overlap passes, the pooled t-test fails to reject, and TOST (Section IV) shows equivalence within 3 points. Vertical lines mark the observed SD quartiles.

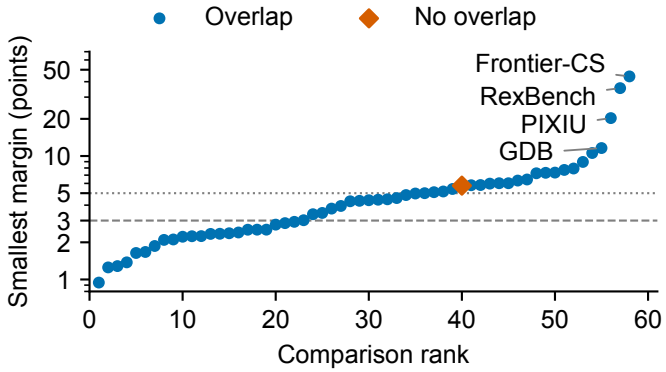


Fig. 2. Smallest equivalence margin supported by each of the 58 comparisons with three runs per side (TOST, 90% interval). The dashed and dotted lines mark 3 and 5 points, respectively. The diamond marks USACO at rank 40. Labels mark the four largest absolute gaps between means. All four pass range overlap.

V. RQ3: A BETTER CHECK AND ITS COST

Runs. Table II gives the runs per side an equivalence test needs for 80% power (an 80% chance of showing equivalence when the true gap is as stated) at the SD quartiles observed in the 58 comparisons (1.1, 1.7 and 2.7 points), for margins of 2, 3 and 5 points, under a true gap of 0 and of 0.6 points (about the observed median). At the median noise a 3-point margin needs about 7 runs per side; per comparison with its own SD the median need is 7 (quartiles 4 and 15.5), and two comparisons would need more than 400. The plug-in rule, the smallest k whose interval from the observed SD just fits the margin, has only 12 to 78% power across this table.

Per-task outcomes. Of 44 adapters with outcomes from both sides, 27 meet the inclusion criteria of replicate runs, at least 20 shared tasks, and validation against published scores or a documented exception. The guide uses per-task checks to diagnose non-overlap [1]. I propose reporting them for every port. With shared per-task pass probabilities, the paired interval should be about a sixth narrower than the run-level

TABLE II
RUNS PER SIDE FOR 80% POWER OF THE POOLED-t TOST (5% LEVEL, NORMAL NOISE, COMMON SD) TO SHOW EQUIVALENCE WITHIN $\pm m$ POINTS AT THE OBSERVED SD QUARTILES, FOR A TRUE GAP OF 0 (AND OF 0.6 POINTS).

margin	SD 1.1	SD 1.7	SD 2.7
±2	7 (9)	14 (20)	33 (49)
±3	4 (4)	7 (8)	16 (18)
±5	3 (3)	3 (4)	6 (7)

interval, which estimates noise from three scores. In fact, the paired interval is wider in 15 of the 26 adapters with a run-level interval (SWE-bench has one original run). Task-level differences are more dispersed than the shared-probability model predicts in 14 of 27 adapters at an unadjusted 5% threshold and in 11 after Benjamini–Hochberg correction at a 5% false discovery rate. A within-task permutation test finds no statistically detectable mean gap in any of the 27 ($p > 0.05$). The stratified Cochran–Mantel–Haenszel test compares the sides within each task and pools evidence across tasks. It finds no mean gap among the 21 adapters with binary outcomes. The task-level differences could come from the port or from side-specific effects such as seeds or caching.

Run-score dispersion. For each of 30 binary adapters with validated replicate Harbor runs, I divide the observed run-score variance by the variance implied by within-task flip rates under independent outcomes. The ratio's quartiles are 0.10, 0.52 and 1.26. Nine adapters have a ratio below 0.1, compared with 2.6 expected under independence ($p = 0.0007$). Low ratios could reflect negative correlations between tasks within a run or dependence among retained runs. The cause remains open.

Injected defects. I simulated both sides of the same 30 adapters. For a task with s passes in r Harbor runs, I use the Beta(0.05, 0.05) posterior mean $(s + 0.05)/(r + 0.1)$. The median run SD is 1.34 points in observed runs using common tasks, 2.25 in the model with this prior, and 3.81 with Beta(1,1). The default is closer to the observed median. Both

modeled medians are higher, consistent with the low dispersion ratios. I break or leak a fraction of port-side tasks by forcing them to fail or pass, with 2,000 replications per cell (Fig. 3). With three runs, a defect that breaks 10% of tasks (a loss of 5.4 points on average) is caught by range overlap 0.64 of the time, at a false-alarm rate of 0.07 under this model; by the pooled t-test 0.59; by the stratified test 0.67; and by a paired t-test on per-task differences 0.54, the last three at 0.05. The paired t suffers because its standard error absorbs the between-task spread of the defect itself, which does not shrink with runs. With ten runs the pooled t and the stratified test catch the same defect 0.86 of the time, the paired t 0.67, and range overlap 0.36. A perfect port is shown equivalent within 5 points 0.51 of the time at $k = 3$, 0.75 at $k = 5$ and 0.94 at $k = 10$. The 21 adapters with a dispersion ratio of at least 0.1 give the same picture; a flat Beta(1,1) prior lowers detection at $k = 3$ to 0.43, 0.34, 0.45 and 0.39 (same order) and the within-5 rate for a perfect port to 0.20; resampling the 8 fractional-score adapters' observed rewards, where the stratified test does not apply, gives 0.72, 0.67 and 0.59 for range overlap, pooled t and paired t.

A verdict tool. The verdict script reads any `parity_experiment.json` and prints the overlap verdict, the gap with its 90% interval, the smallest provable margin, and the runs per side needed for 80% power at that row's own noise if the true gap were 0. For USACO: gap $+2.6$ (-0.5 to $+5.8$), margin 5.8 points, 7 runs for ± 3. For RexBench: gap $+8.3$ (-18.8 to $+35.5$), margin 35.5 points, 464 runs.

VI. Recommended protocol

1) State the margin before running: how many points of difference would make the port a different benchmark (3 and 5 points are illustrative; justify the margin for the benchmark).
2) Choose the run count from the noise for 80% power at that margin (Table II); when the noise is unknown, start with at least five runs per side and fix the total from their SD before looking at the gap. If the runs needed are out of reach (464 per side for RexBench at ± 3), enlarge the parity set where a subset was used, or report the smallest provable margin instead.
3) Report the gap with its 90% interval and the smallest provable margin. Declare "equivalent within $\pm m$" only when the interval lies inside $\pm m$; otherwise "not yet shown", which asks for a larger experiment, not a rewrite.
4) Publish aligned per-task outcomes for both sides, report a stratified-by-task test (for binary outcomes; the pooled t on run scores otherwise), and investigate tasks that differ in every run, which the guide's playbook already treats as a sign of a systematic issue [1].
5) Keep range overlap as a sanity check at most.

The cost is about seven runs per side instead of three at the median observed noise for a 3-point margin, and a reporting change.

VII. Threats to validity

Noise models. Run scores are treated as independent draws with a common variance; binomial and per-task models rank the rules the same way. Run-level shocks such as infrastructure noise [6] are not modelled. *Selection.* Published pass rates can reflect which results are retained (Section IV). *Few runs.* Three runs give a noisy SD; per-task data give a second estimate. *Metric rule.* The main-metric rule and its amendments are a choice; Section IV reports how the within-5 count moves under a 0.1-point shift and under Welch intervals. Mixed metric types sit on one point scale; subsets are noisier than full sets. *Per-task data.* One format per adapter, validated only against the published run scores; 13 of the 44 two-sided folders are excluded, 4 are Terminal-Bench legs reported in the repository only, and 6 of the 27 used carry documented exceptions. The per-task model is sensitive to the prior and assumes independent task outcomes. Run-level intervals can be too narrow if the retained runs understate variability across independent repetitions. *Snapshot.* One commit of a moving repository; no benchmark was rerun.

VIII. Related work

That overlap of error bars or intervals is a poorly calibrated test is classical [11], [12], and researchers misjudge it even so [13]. Equivalence testing with a stated margin is standard in other fields [9], [10]. Noise in language-model and agent evaluation has been quantified with error bars and clustering [4], variance decomposition [5] and controlled infrastructure experiments [6]; re-implementations of the same benchmark are known to drift [14], and other ecosystems accept re-implementations with a fixed tolerance [15] or a test against a reference distribution [16]. Recent work applies paired per-task tests and TOST at a stated margin to agent scaffolds and compressed models [17], [18], while a leaderboard audit uses paired tests and states no equivalence margin [19]; Zhang *et al.* distinguish a check that fails to detect harm from one that certifies non-inferiority [20]. Inside Harbor, the maintainers' paper reports a Welch t-test and a permutation test for two adapters [3], and one port that had passed parity with one model later proved to leak gold tests and was fixed the same day [21]. I focus on the acceptance rule itself, the evidence in the published parity runs, and the run budget needed for an equivalence check.

IX. Conclusion

With three runs per side and exchangeable scores under parity, range overlap has a false-alarm rate of at most 10%; a pass does not certify closeness. Of the 58 published comparisons, 36 show equivalence within 5 points. For a 3-point margin at the median observed SD and a true gap of zero, about seven runs per side provide 80% power. Stated margins, equivalence tests and aligned per-task outcomes make the check more informative.

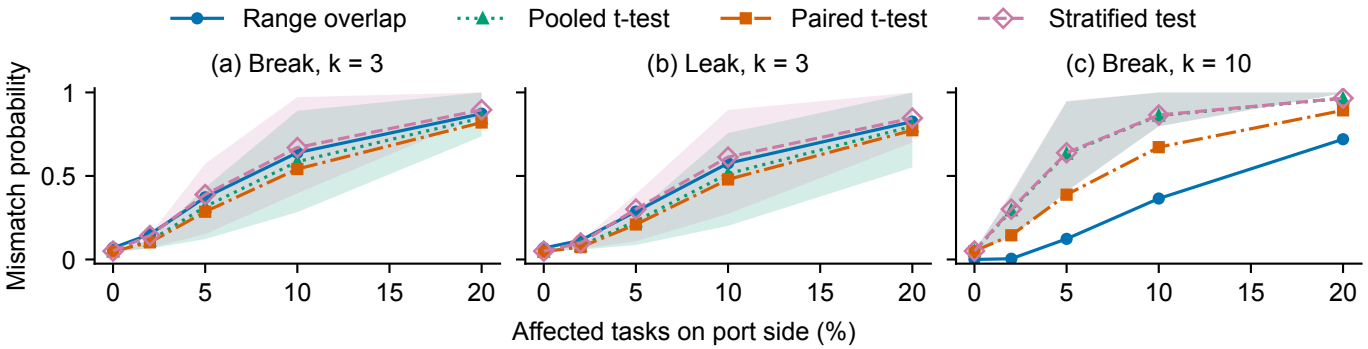


Fig. 3. Detection of injected defects, mean over 30 adapters simulated from their per-task outcomes: (a) break defects at $k = 3$, (b) leak defects at $k = 3$, (c) break defects at $k = 10$. Bands: interquartile range across adapters for the pooled t and stratified tests; in (c) their curves nearly coincide. The value at zero is the false-alarm rate.

Acknowledgment

I thank the Harbor maintainers for publishing their runs and per-task files.

Under my direction, Claude (Anthropic) wrote most of the analysis code for Sections III–V, ran the literature search for Section VIII and drafted the text, which I revised. Codex (OpenAI) corrected grammar and revised the plotting code for Figs. 1–3.

References

[1] Harbor Framework contributors, "Adapters (agent guide)," https://github.com/harbor-framework/harbor/blob/dc933be/adapters/ADAPTER_CONTRIBUTING.md, 2026.

[2] M. A. Merrill, A. G. Shaw, N. Carlini *et al.*, "Terminal-Bench: Benchmarking agents on hard, realistic tasks in command line interfaces," arXiv:2601.11868v1, 2026.

[3] L. Shi, H. Lin, Z. Zhu *et al.*, "Harbor adapters and Harbor-Index: Infrastructure and a curated meta-dataset for large-scale agentic evaluation," arXiv:2609.04298v3, 2026.

[4] E. Miller, "Adding error bars to evals: A statistical approach to language model evaluations," arXiv:2411.00640v1, 2024.

[5] M. Hardy, R. Azam, A. Reuel, M. Kochenderfer, and S. Koyejo, "Agent evaluation reliability: More tasks won't (always) fix an agent leaderboard," arXiv:2610.00651v1, 2026.

[6] G. Segato, "Quantifying infrastructure noise in agentic coding evals," Anthropic Engineering, February 2026. https://www.anthropic.com/engineering/infrastructure-noise, 2026.

[7] Harbor Framework contributors, "Split adapter tutorial to human/AI, update registry handling," Pull request #1403, https://github.com/harbor-framework/harbor/pull/1403, 2026.

[8] ——, "parity-experiments (Hugging Face dataset)," https://huggingface.co/datasets/harborframework/parity-experiments, 2026, accessed 11 Oct. 2026.

[9] D. J. Schuirmann, "A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability," *Journal of Pharmacokinetics and Biopharmaceutics*, vol. 15, no. 6, pp. 657–680, 1987.

[10] D. Lakens, "Equivalence tests: A practical primer for t tests, correlations, and meta-analyses," *Social Psychological and Personality Science*, vol. 8, no. 4, pp. 355–362, 2017.

[11] N. Schenker and J. F. Gentleman, "On judging the significance of differences by examining the overlap between confidence intervals," *The American Statistician*, vol. 55, no. 3, pp. 182–186, 2001.

[12] M. E. Payton, M. H. Greenstone, and N. Schenker, "Overlapping confidence intervals or standard error intervals: What do they mean in terms of statistical significance?" *Journal of Insect Science*, vol. 3, no. 1, p. 34, 2003.

[13] S. Belia, F. Fidler, J. Williams, and G. Cumming, "Researchers misunderstand confidence intervals and standard error bars," *Psychological Methods*, vol. 10, no. 4, pp. 389–396, 2005.

[14] S. Biderman, H. Schoelkopf, L. Sutawika *et al.*, "Lessons from the trenches on reproducible evaluation of language models," arXiv:2405.14782, version 3 (2026), 2024.

[15] A. Abbas, C. Waggoner, and J. Olive, "Developing and maintaining an open-source repository of AI evaluations: Challenges and insights," arXiv:2507.06893v1, 2025.

[16] MLCommons, "MLPerf training rules," https://github.com/mlcommons/training_policies/blob/master/training_rules.adoc, 2026, commit 3828931.

[17] Y. Liu and Z. Han, "What does a harness buy? Tokens, mostly," arXiv:2610.04433v1, 2026.

[18] A. Singh, "Certifying compressed language models: An audit and a statistical toolkit," arXiv:2608.15046v1, 2026.

[19] F. Liu, Y. Liu, R. Sun, L. Luo, and S. Guo, "Coding agents have converged: Why the SWE-bench leaderboard can no longer order its top entries, and what to measure instead," arXiv:2609.17394v1, 2026, to appear at ADMA 2026.

[20] S. Zhang, T. Wu, and F. Qian, "Certifying model upgrades with slice-wise non-regression and incumbent fallback," arXiv:2609.13714v1, 2026.

[21] Harbor Framework, "Claude 4.5+ & gpt5.4 models doing too well on Swebenchpro," https://github.com/harbor-framework/harbor/issues/1067, 2026.